

An OpenSCAD-supported Tool for Prototyping Functional Solutions in Robotic Design

ROLLE, Guillermo; NACUSSE, Matías; JUNCO, Sergio

Laboratorio de Control y Automatización (LAC), FCEIA, UNR

Abstract

The development of robotic systems demands that experts from multiple engineering domains (ranging from Mechanical or Electronics to Control, Software or Systems) share a common language for exchanging information and design decisions efficiently. In this regard, when a set of conceptual solutions needs to be evaluated, engineers from various domains are required to sketch, prototype and evaluate different alternatives altogether which, in turn, may impose constraints over other engineering domains.

This paper presents a tool that enables engineers to generate conceptual robot prototypes to be simulated with ROS and Gazebo by using OpenSCAD to quickly sketch 3D primitive-based parts and assemblies from a coding interface. The tool uses a custom-made library that allows users to connect mechanical components via the definition of joints, mirroring the structure of URDF files. A custom Python script processes the OpenSCAD model to: (1) automatically export individual STL meshes for every component; (2) compute mass, center of mass and inertia tensors; and (3) generate a complete URDF description in the xacro format split into the standard description and gazebo packages.

The workflow is integrated into an MBSE environment using ARCADIA/Capella. Architectural choices captured in Capella are directly reflected in the OpenSCAD model, ensuring consistency between logical architecture and the physical prototype. By treating geometry as code, the tool accelerates design iterations and variants evaluation by reducing the workload required to generate a virtual prototype from a functional concept.

We demonstrate the effectiveness of the approach through a representative case study of a robotic manipulator, showing that different alternatives can be tested in simulation faster compared to raw modification of URDF files or generation from other CAD tool.

Keywords

Mechatronic Design, Model-Based Systems Engineering (MBSE), OpenSCAD, ROS/Gazebo